

COURSE TITLE : **COMPUTER HARDWARE & NETWORKING LAB**
 COURSE CODE : **4267**
 COURSE CATEGORY : **A**
 PERIODS/WEEK : **6**
 PERIODS/SEMESTER : **90**
 CREDITS : **3**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Identification of PC components and PC assembling	22
2	Basic trouble shooting of PC and OS installation	23
3	LAN installation and Crimping of LAN cable	22
4	Configure Network settings in OS and simple trouble shooting	23

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able to :
1	1	To assemblea Personal Computer System
2	1	To trouble shoot the problems of PC and Install OS
3	1	To demonstrate LAN installation and Crimping of LAN cable
4	1	To Configure Network settings in OS and simple trouble shooting

Specific Outcomes:

MODULE – I: Identification of PC components and PC assembling

1.1 To assemblea Personal Computer System

- 1.1.1 List PC components
- 1.1.2 Identify PC components
- 1.1.3 Identify its ports and sockets
- 1.1.4 Demonstrate SMPS
- 1.1.5 DemonstrateMotherboard
- 1.1.6 Demonstrate how to assemble a PC

MODULE – II: Basic trouble shooting of PC and OS installation

- 2.1 To trouble shoot the problems of PC and Install OS
 - 2.1.1 Identify the beep sound produces by the motherboard
 - 2.1.2 Identify the troubles in PC
 - 2.1.3 Solve the simple problems in PC
 - 2.1.4 Install Linux in a newly assembled PC
 - 2.1.5 Show hard disk partitioning and formatting
 - 2.1.6 Demonstrate hard disk file system
 - 2.1.7 Install application software in Linux

MODULE – III: LAN installation and Crimping of LAN cable

3.1 To demonstrate LAN installation and Crimping of LAN cable

- 3.1.1 Demonstrate the structure of LAN
- 3.1.2 Identify the components in LAN
- 3.1.3 Demonstrate the Ethernet Switch
- 3.1.4 Demonstrate NIC card
- 3.1.5 Demonstrate the colour code of LAN cable and Crimping
- 3.1.6 Demonstrate how to setup a LAN using Switch and PC

MODULE – IV: Configure Network settings in OS and simple trouble shooting

4.1 To Configure Network settings in OS and simple trouble shooting

- 4.1.1 Demonstrate Linux network connection settings
- 4.1.2 Use commands like ping, ifconfig etc.
- 4.1.3 Demonstrate about IP address and MAC address
- 4.1.4 Demonstrate about static and dynamic IP address settings
- 4.1.5 Demonstrate about DNS and Gateway settings
- 4.1.6 Identify simple problems in LAN
- 4.1.7 Solve the simple problems in LAN